

Format of Question paper for Midsemester Examination

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2022-2023

Comprehensive Examination
(EC-2 Makeup)

Course No. : AIMLCL ZG517
Course Title : FAccT Machine Learning
Nature of Exam : Open Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : 5th July 2023

No. of Pages	= 3
No. of Questions	= 3

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1.

[3+3+3+2=11 Marks]

- A. Consider gender debiasing regularization for word embeddings. The following are the word embeddings. $w_g(.)$ and $w_a(.)$ are the gender-related and gender-neutral parts.

Word	Male seed words	Female seed words	$w_g(1)$	$w_g(2)$	$w_g(3)$	$w_a(1)$	$w_a(2)$
daughter		X	0.3	0.5	0.6	0.3	0.6
son	X		.75	0.6	0.3	0.6	0.5
prince	X		0.75	0.6	0.3	0.3	0.5
princess		X	0.3	0.5	0.6	0.75	0.6
doctor			0.4	0.3	0.8	0.8	0.1
lawyer			0.2	0.1	0.8	0.4	0.3

- i) Calculate the gendered word regulating component J_D .

$$J_D = - \left\| \sum_{w \in \Omega_M} w^{(g)} - \sum_{w \in \Omega_F} w^{(g)} \right\|_1 + \sum_{w \in \Omega_M} \left\| 1 - w^{(g)} \right\|_2^2 + \sum_{w \in \Omega_F} \left\| -1 - w^{(g)} \right\|_2^2$$

$$\begin{aligned} J_D &= \left\| [.75*2 \ 0.6*2 \ 0.3*2] - [0.3*2 \ 0.5*2 \ 0.6*2] \right\|_1 \\ &\quad + 2 \left\| [1-0.75 \ 1-0.6 \ 1-0.3] \right\|_2^2 + 2 \left\| [-1-0.3 \ -1-0.5 \ -1-0.6] \right\|_2^2 \\ &= 1.7+1.425+13=16.125 \end{aligned}$$

- ii) Calculate the gender-neutral word regulating component J_E .

$$J_E = \sum_{w \in \Omega_N} \left(v_g^T w^{(a)} \right)^2$$

and

$$v_g = \frac{1}{|\Omega'|} \sum_{(w_m, w_f) \in \Omega_M, \Omega_F} (w_m^{(a)} - w_f^{(a)})$$

$$v_g = \frac{1}{4} * ([0.6 \ 0.5] - [.3 \ .6] + [0.6 \ 0.5] - [.75 \ .6] + [0.3 \ 0.5] - [.3 \ .6] + [0.3 \ 0.5] - [.75 \ .6]) \\ = \frac{1}{4} * [-0.3 \ 0.4]$$

$$J_E = \frac{1}{4} * (-.8 * .3 + 0.1 * 0.4 - 0.4 * 0.3 + 0.3 * 0.4) = 0.05$$

- B. Apply universal sampling to generate a training dataset from the following dataset. Gender is the protected attribute, P is the predictor and C is the target class. How many distinct datasets can be created?

Gender	P	C
F	6	-
F	8	-
F	9	-
F	11	+
M	9.5	-
M	12	+
M	13	+
M	16.5	+

One possible training set after sampling will be

Gender	P	C
F	8	-
F	9	-
F	11	+
F	11	+
M	9.5	-
M	9.5	-
M	12	+
M	13	+

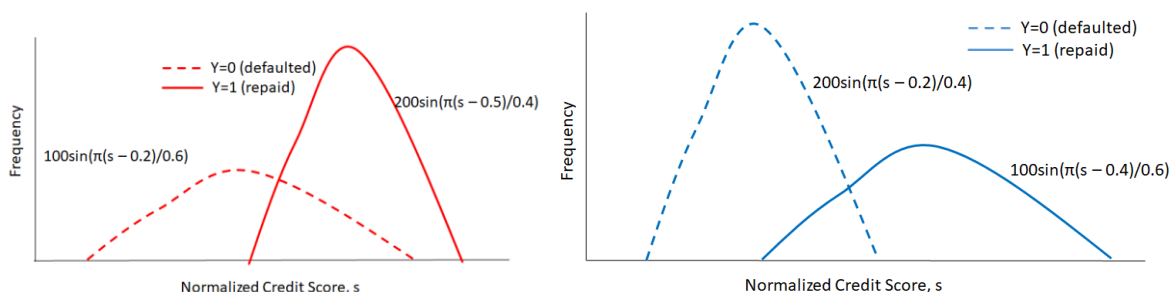
(F, -) and (M, +) need to be undersampled from 3 (in the original dataset) to 2 instances (after sampling). Since there are 3 instances of (F, -) and (M, +) in the original set, total number of distinct training set that can be created will be $3 \times 3 = 9$. The dataset shown above is one such training set.

Q.2. Consider a financial institution that uses normalized credit score for approving or rejecting housing loan. Note there are two population of applicants 'blue' (deprived group with protected attribute $A=0$) and 'red' (favored group with $A=1$). Loan is approved, i.e., $y'=1$ if normalized credit score is greater than some threshold t , else rejected. $y'=1$ indicates approved loans that are repaid based on historical data, and $y=0$ indicates approved loan was

defaulted. The financial institution wants to approve loans that is likely to be repaid. Distributions are described in following figure (not drawn to scale) for both populations.

Note: $\int \sin(\alpha x) dx = -\cos(\alpha x)/\alpha$

[2+4+4=10 Marks]



- A. Calculate probability $P(y'=1)$ for both 'blue' and 'red' populations for threshold $t=0.4$.

For blue, $P(y'=1) = (100+100)/300=0.67$

For red, $P(y'=1)=100(1+\cos(\pi/3))+200*2 /600 = 0.916$

- B. If threshold $t=0.4$ is chosen for the blue population, what threshold value, if any for the red population will ensure demographic parity?

At $t=0.4$, for blue, $P(y'=1)=.67$

Let at $t=t_{red}$ $P(y')=0.67$

Then, $200(1+\cos(\pi(t_{red}-0.5)/0.4)) + 100(1+ \cos(\pi(t_{red} - 0.2)/0.6)) = 400$

Or, $2 \cos(\pi(t_{red}-0.5)/0.4) + \cos(\pi(t_{red} - 0.2)/0.6) = 1$

Or, $t_{red} = 0.59435$

- C. If threshold $t=0.6$ is chosen for the blue population, calculate the probability $P(y'=1|y=1)$. What threshold value for the red population will ensure equal opportunity? For the same threshold value, calculated the probability of $P(y'=1|y=0)$ for both blue and red populations. Is equality of odds achieved?

At $t=0.6$, $P(y'=1|y=1) = (1+\cos(\pi/3))/2=0.75$ for blue

For red, $t=0.6534$ $P(y'=1|y=1) = 0.75$

[complete the $P(y'=1|y=0)$ for both blue and red and compare]

Q.3. Consider the following training set. Gender is the sensitive attribute, Skill and Years are predictors and Y is the target decision.

[2+2+5=9 marks]

Gender	Skill	Years of Experience	Promotion Decision Y
Female	Java	2	no
Female	C++	5	yes
Male	Java	2	yes
Male	C++	3	yes

A. Calculate the Prejudice Index (PI) and Normalized Prejudice Index (NPI) of the dataset.

$$PI = \sum_{(y,s) \in \mathcal{D}} \hat{Pr}[y, s] \ln \frac{\hat{Pr}[y, s]}{\hat{Pr}[y] \hat{Pr}[s]}$$

$$NPI = PI / (\sqrt{H(Y)H(S)})$$

$$\begin{aligned} Pr[-,F] &= 1/4 & Pr[+,F] &= 3/4 & Pr[-,M] &= 0 & Pr[+,M] &= 1/2 \\ Pr[-] &= 0.25 & Pr[+] &= 0.75 & Pr[F] &= 0.5 & Pr[M] &= 0.5 \end{aligned}$$

$$\begin{aligned} \text{So, } PI &= \frac{1}{4} \ln(3/4/0.75*0.5) + \frac{1}{4} \ln(1/4/0.25*0.5) + \frac{1}{2} \ln(1/2/0.75*0.5) \\ &= \frac{1}{4} \ln(2) + \frac{1}{4} \ln(2) + \frac{1}{2} \ln(4/3) \end{aligned}$$

$$H(S)=1 \quad H(Y)=1/4 \ln(1/4)+3/4 \ln(3/4), \text{ so } NPI = PI$$

B. A logistic regression based classifier with prejudice removal regularizer is trained to obtain fair decisions. All classifier parameters except the one associated with Gender are initialized to 0. Parameter w_{gender} associated with Gender is initialized to 1. Female and C++ are encoded as 0 and Male and Java are encoded as 1 while training the classifier. Assuming learning rate = 0.1 and regularization constant is 0.5, calculate the value of classifier parameters after 1st iteration without and with prejudice removal regularizer. Assume batch training is being used. Show all calculation steps.

Gender	Skill	Years of Experience	Promotion Decision Y	initial Output $\mathcal{M}$
0	1	2	no	0.5
0	0	5	yes	0.5
1	1	2	yes	$1/(1+e^{-1})$ =0.731
1	0	3	yes	$1/(1+e^{-1})$ =0.731

So, in the next iteration without regularization,

$$w_0 = 0 - 0.1/4 * ((0.5) + (0.5-1) + (1-0.731) + (1-0.731)) = 0.013$$

$$w_{\text{gender}} = 1 - 0.1/4 * ((1-0.731) + (1-0.731)) = 1.013$$

$$w_{\text{skill}} = 0 - 0.1/4 * (0.5 + (1-0.731)) = -0.006$$

$$w_{\text{exp}} = 0 - 0.1/4 * (0.5*2 + (0.5-1)*5 + (0.731-1)*2 + (0.731-1)*3) = 0.071$$

If PI removal regularization is used, regularization loss R to be optimized is given by

$$\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \Theta] \ln \frac{\hat{Pr}[y|s_i]}{\hat{Pr}[y]}$$

$$\hat{Pr}[y] = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \mathcal{M}[y|\mathbf{x}_i, s_i; \Theta]$$

$$\hat{\Pr}[y|s] \approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s} \mathcal{M}[y|\mathbf{x}_i, s; \boldsymbol{\Theta}]}{|\{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s\}|}$$

Gender	Skill	Years of Experience	Promotion Decision Y	initial Output $\mathcal{M}$	$\ln(\Pr[y s_i]/\Pr[y])$
0	1	2	no	0.5	$\ln(0.5/0.6155)=-0.21$
0	0	5	yes	0.5	$\ln(0.5/0.6155)=-0.21$
1	1	2	yes	$1/(1+e^{-1})=0.731$	$\ln(0.731/0.6155)=.17$
1	0	3	yes	$1/(1+e^{-1})=0.731$	$\ln(0.731/0.6155)=0.17$

After regularization is applied,

$$w_0 = 0.013 - 0.5 * 0.1 / 4 * ((0.5) * (-0.21) + (0.5 - 1) * (-0.21) + (0.731 - 1) * 0.17 + (0.731 - 1) * 0.17) = 0.0146$$

$$w_{\text{gender}} = 1.013 - 0.5 * 0.1 / 4 * ((0.5) * (-0.21) * 0 + (0.5 - 1) * (-0.21) * 0 + (0.731 - 1) * 0.17 + (0.731 - 1) * 0.17) = 1.013$$

Similarly, $w_{\text{skill}} = -0.0062$ and $w_{\text{exp}} = 0.072$